# Proving angle relationships

**1** The rays $\overrightarrow{DE}$ and $\overrightarrow{DF}$ are opposite rays.
Determine whether the following are true statements.

**a** $\overrightarrow{DE}$ and $\overrightarrow{DF}$ are the legs of a straight angle.

**b** $m\angle EDF = 180^\circ$

**c** $\overrightarrow{DE}$ and $\overrightarrow{DF}$ are the legs of a right angle.

**d** $\overrightarrow{DE}$ and $\overrightarrow{DF}$ are not the legs of any angle.

**e** $m\angle EDF = 0^\circ$

**2** The rays $\overrightarrow{DE}$ and $\overrightarrow{DF}$ are perpendicular to one another.
Determine whether the following statements are true.

**a** $\angle EDF$ is a right angle.

**b** $m\angle EDF = 180^\circ$

**c** $m\angle EDF = 90^\circ$

**d** $\angle EDF$ is a straight angle.

**3** Given that two angles form a linear pair, determine whether the following statements are always true.

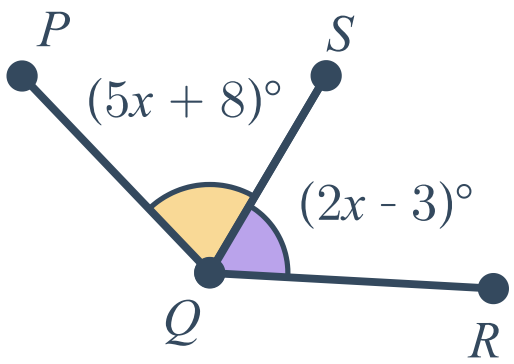
**a** They are complementary.

**b** They are congruent.

**c** They are adjacent.

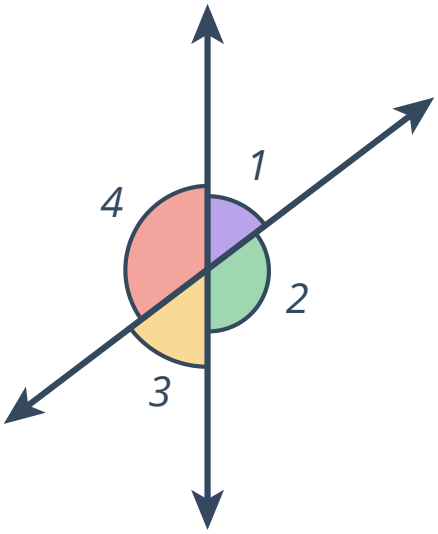
**d** They are supplementary.

**4** Consider the attached diagram.
Solve for $x$, given that $m\angle PQR = 145^\circ$.
Justify each steps using a two column proof.




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**5** Without directly using the vertical angles congruence theorem, construct a two column proof to prove that $\angle 1 \cong \angle 3$.

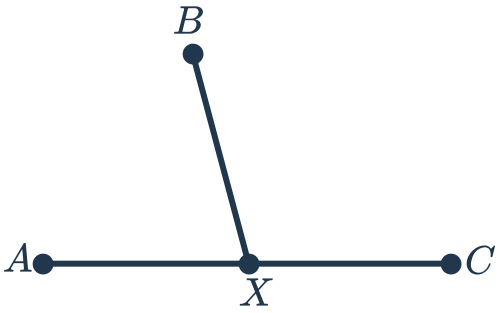



## Additional questions

**6** We are given that the two marked angles are congruent.
Construct a two column proof to solve for $x$.




**7** Write a two column proof to prove the linear pairs postulate.
Given: $\angle AXB$ and $\angle BXC$ are a linear pair
Prove: $\angle AXB$ and $\angle BXC$ are supplementary

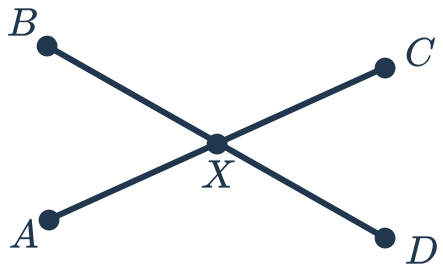



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**8** Write a two column proof to prove the vertical angles theorem.
Given:

- $\overline{AC}$ is a straight line segment
- $\overline{BD}$ is a straight line segment

Prove: $\angle AXB \cong \angle CXD$




**9** Construct a two column proof to prove the congruent complements theorem.
Given:

- $\angle A$ and $\angle C$ are complementary
- $\angle B$ and $\angle C$ are complementary

Prove: $\angle A \cong \angle B$

**10** Construct a two column proof for the theorem "all right angles are congruent".
Given: $\angle A$ and $\angle B$ are right angles
Prove: $\angle A \cong \angle B$

**11** Write a two column proof for the theorem "if two angles are congruent and supplementary, both are right angles".
Given: $\angle A$ and $\angle B$ are congruent and supplementary
Prove: $\angle A$ and $\angle B$ are right angles

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